DSA Assignment 4 Report

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Section- A

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Data Management Mini Toolkit (DMMT)

Data Structures Assignment Report - Unit 4

1. Binary Search Tree (BST)

The BST is a hierarchical data structure where each node has at most two children. The left child contains values smaller than the parent, and the right child contains values larger.

Why Inorder Traversal is Sorted

Inorder traversal visits nodes in the order: Left -> Root -> Right. Because of the BST property (Left < Root < Right), this recursive visit pattern naturally picks up the smallest elements first and the largest elements last, resulting in a strictly increasing sequence.

Time Complexity Analysis

Scenario	Complexity	Description
Average Case	O(log n)	When the tree is balanced, each step halves the search space.
Worst Case	O(n)	Occurs when the tree is skewed (like a linked list).

2. Graph Representation & Traversal

The toolkit implements a directed, weighted graph using an Adjacency List. This is more memory-efficient than a matrix for sparse graphs.

BFS vs. DFS

- Breadth-First Search (BFS): Uses a Queue. It explores neighbors level by level. It is ideal for finding the shortest path in unweighted graphs.
- Depth-First Search (DFS): Uses a Stack (or recursion). It explores as far as possible along each branch before backtracking. It is excellent for cycle detection and pathfinding.

Real-World Use Cases

- BFS: Used by GPS systems to find the fewest number of hops between two locations.
- DFS: Used in compilers for dependency analysis and topological sorting.

3. Hash Table with Separate Chaining

Hashing maps large keys to small indices for O(1) average lookup time.

Collisions and Resolution

A collision occurs when two different keys result in the same index after the hash function (key % size). We resolved this using Separate Chaining, where each index in the table points to a list (chain) of entries.

Time Complexity

- Average Case: O(1) - When keys are distributed uniformly.
- Worst Case: O(n) - When all keys hash to the same index, turning the table into a linked list.